**Johannes Kepler Universität Linz**
Michael Neunteufel

**Numerical Methods for Elliptic Problems – Exercise Sheet 3**

Due date: November 10, 2026, 10:15-11:45, Room SP2-0346

**Exercise 16 (Trace space):**
Let $Q = (0,1)^2$ and identify $\Gamma = (0,1) \times \{0\}$ with $(0,1)$. Set

$$e_0(x) = 1, \qquad e_n(x) = \sqrt{2}\cos(n\pi x) \quad (n \geq 1), \qquad \lambda_n = 1 + n^2\pi^2, \quad \mu_n = \sqrt{\lambda_n}.$$

Let $W := \gamma_\Gamma\big(H^1(Q)\big)$ carry the quotient norm $\|g\|_W := \inf_{\gamma_\Gamma u = g} \|u\|_{H^1(Q)}$. You may use that $(e_n)_{n\geq 0}$ is an orthonormal basis of $L^2(\Gamma)$ and the independent Fourier definition for $0 \leq s \leq 1$

$$H^s(\Gamma) := \left\{ g = \sum_{n\geq 0} a_n e_n \in L^2(\Gamma) : \sum_{n\geq 0} \lambda_n^s |a_n|^2 < \infty \right\},$$

with squared Fourier norm $\sum_n \lambda_n^s |a_n|^2$. The following facts may be used without proof. For $u \in H^1(Q)$, $u_n(y) := \int_0^1 u(x,y)e_n(x)\,dx$ belong to $H^1(0,1)$, $u_n(0) = (\gamma_\Gamma u, e_n)_{L^2(\Gamma)}$, and

$$\|u\|^2_{H^1(Q)} = \sum_{n\geq 0} \int_0^1 \big(|u_n'|^2 + \mu_n^2 |u_n|^2\big)\,dy \qquad \text{(Parseval)}.$$

For $v \in H^1(0,1)$ and $\mu \geq 1$,

$$\mu|v(0)|^2 \leq 2\int_0^1 \big(|v'|^2 + \mu^2|v|^2\big)\,dy \qquad \text{(1D trace estimate)}.$$

**a)** For $u \in H^1(Q)$ with trace $g = \sum_{n\geq 0} a_n e_n$, prove

$$\sum_{n\geq 0} \mu_n |a_n|^2 \leq 2\|u\|^2_{H^1(Q)}.$$

Deduce a lower bound for $\|g\|^2_W$.
**b)** Let $\sum_{n\geq 0} \mu_n |a_n|^2 < \infty$ and set $U_N(x,y) := \sum_{n=0}^N a_n e_n(x) e^{-\mu_n y}$. Use orthogonality to prove

$$\|U_N\|^2_{H^1(Q)} = \sum_{n=0}^N \mu_n (1 - e^{-2\mu_n})|a_n|^2 \leq \sum_{n\geq 0} \mu_n |a_n|^2.$$

Show that $U_N$ converges in $H^1(Q)$ to a function with trace $g = \sum_{n\geq 0} a_n e_n$. *Hint:* Apply the same identity to $U_N - U_M$, and use completeness and continuity of $\gamma_\Gamma : H^1(Q) \to L^2(\Gamma)$.
**c)** Conclude that $W = H^{1/2}(\Gamma)$ with equivalent quotient and Fourier norms. Compare the squared-norm weights for $L^2(\Gamma)$, $H^{1/2}(\Gamma)$, and $H^1(\Gamma)$ and explain the exponent $1/2$.

**Exercise 17 (Shape regularity):**
Let $\mathcal{F}$ be a family of nondegenerate triangles in $\mathbb{R}^2$. For $T \in \mathcal{F}$, let $h_T = \operatorname{diam}(T)$, $\rho_T$ its inradius, $\alpha_T$ its smallest interior angle, and $|T|$ its area.

**a)** Show that the following conditions are equivalent, with constants independent of $T \in \mathcal{F}$

$$\text{(i) } \rho_T \geq \gamma h_T, \qquad \text{(ii) } \alpha_T \geq \alpha_0, \qquad \text{(iii) } |T| \geq c h_T^2,$$

for some positive constants $\gamma, \alpha_0, c$.

**b)** Does a uniform bound on the ratio of the longest to the shortest edge imply shape regularity? Give a proof or a counterexample.

*Hint:* If the side lengths are $a \leq b \leq h_T$ and $s = (a + b + h_T)/2$, use $|T| = \rho_T s$ and $|T| = \frac{1}{2} b h_T \sin \alpha_T$.

**Exercise 18 (A patch-average quasi-interpolant):**
Let $\mathcal{T}_h$ be a conforming, shape-regular triangulation of a bounded Lipschitz polygon $\Omega \subset \mathbb{R}^2$. Let $\mathcal{N}_h$ contain all vertices, including boundary vertices. For $z \in \mathcal{N}_h$, let $\phi_z$ be the corresponding hat function and $\omega_z$ the union of all triangles containing $z$. For $T \in \mathcal{T}_h$, set $\omega_T = \bigcup_{z \text{ vertex of } T} \omega_z$. Consider the Clément-type averaging operator

$$J_h u = \sum_{z \in \mathcal{N}_h} u_z \phi_z, \qquad u_z = \frac{1}{|\omega_z|} \int_{\omega_z} u \, dx.$$

**a)** Show that $J_h$ is defined on $L^2(\Omega)$, reproduces constants, and $(J_h u)|_T$ depends only on $u|_{\omega_T}$. Prove also the local stability estimate

$$\|J_h w\|_{L^2(T)} \leq C \|w\|_{L^2(\omega_T)} \qquad (w \in L^2(\Omega)).$$

*Hint:* Use Cauchy–Schwarz and the nonnegative partition of unity on $T$.

You may use the gradient stability estimate without proof

$$\|\nabla J_h w\|_{L^2(T)} \leq C h_T^{-1} \|w\|_{L^2(\omega_T)} \qquad (w \in L^2(\Omega)),$$

and the patch Poincaré inequality

$$\|u - u_{\omega_T}\|_{L^2(\omega_T)} \leq C h_T \|\nabla u\|_{L^2(\omega_T)}, \qquad u_{\omega_T} := \frac{1}{|\omega_T|} \int_{\omega_T} u \, dx.$$

The constants are independent of the element sizes.

**b)** For $u \in H^1(\Omega)$ prove

$$\|u - J_h u\|_{L^2(T)} \leq C h_T \|\nabla u\|_{L^2(\omega_T)}, \qquad \|\nabla J_h u\|_{L^2(T)} \leq C \|\nabla u\|_{L^2(\omega_T)}.$$

*Hint:* Subtract the same constant $u_{\omega_T}$ on the whole patch.

**Exercise 19 (A quadratic element):**
Let $T$ be a triangle with vertices $z_i$, barycentric coordinates $\lambda_i$ $(i = 1, 2, 3)$, and edges $E_{ij} = [z_i, z_j]$. On the quadratic polynomials $P_2(T)$ consider the degrees of freedom

$$d_i(v) = v(z_i) \quad (i = 1, 2, 3), \qquad m_{ij}(v) = \frac{1}{|E_{ij}|} \int_{E_{ij}} v \, ds \quad (1 \le i < j \le 3).$$

These are *unisolvent* if they uniquely determine every $v \in P_2(T)$.
**a)** Count the degrees of freedom and prove unisolvence. *Hint:* Write $v$ as a linear combination of $\lambda_i^2$ and $\lambda_i \lambda_j$ $(i < j)$.
**b)** Find the basis dual to these six degrees of freedom in barycentric coordinates. *Hint:* Try $\phi_i = A\lambda_i^2 + B\lambda_i$ for a vertex function and $\phi_{ij} = C\lambda_i\lambda_j$ for an edge function.

**Exercise 20 (Volume loads beyond $L^2$):**
Let $\Omega \subset \mathbb{R}^3$ be a bounded Lipschitz domain. Consider $-\Delta u + u = f$ in $\Omega$ with $u = 0$ on $\partial\Omega$.
**a)** Formulate the weak problem on $H_0^1(\Omega)$ and prove existence, uniqueness, and the estimate

$$\|u\|_{H^1(\Omega)} \le C \|f\|_{L^{6/5}(\Omega)} \qquad \text{for every } f \in L^{6/5}(\Omega).$$

Use $H^{-1}(\Omega) = H_0^1(\Omega)^*$ and $\|v\|_{L^6(\Omega)} \le C_S \|v\|_{H^1(\Omega)}$.

Explain briefly why the same existence result holds for $f \in L^p(\Omega)$ with $6/5 \le p \le \infty$.
**b)** Take $\Omega = B_1(0) \subset \mathbb{R}^3$ and $f(x) = |x|^\alpha$. For which $\alpha \in \mathbb{R}$ does $f$ belong to $L^{6/5}(\Omega)$, and for which $\alpha$ to $L^2(\Omega)$? Give an explicit admissible load which is not in $L^2(\Omega)$.

**Exercise 21:**
Consider the conducting disk $B := B_{1/4}((0, 1))$ above the grounded line $y = 0$. Its potential is one. For $R > 5/4$, truncate the upper half-plane to $\Omega_R := ((-R, R) \times (0, R)) \setminus \overline{B}$. Let $u_D^R, u_N^R$ be harmonic in $\Omega_R$, equal one on $\partial B$ and zero on the bottom edge. On the other three outer edges impose, respectively, $u_D^R = 0$ and $\partial_n u_N^R = 0$. Define the corresponding energies as

$$E_\sigma(R) := \int_{\Omega_R} |\nabla u_\sigma^R|^2 \, dx, \qquad \sigma \in \{D, N\}.$$

**a)** Characterize both solutions as energy minimizers (define its space of admissible functions). Let $E_\infty$ be the minimum exterior gradient energy with the same conductor/ground data and decay at infinity. Use restriction and zero extension to show that $E_N(R)$ $(E_D(R))$ is monotone increasing (decreasing), and

$$E_N(R) \le E_\infty \le E_D(R).$$

**b)** Use `exterior_electrostatics.ipynb` to plot both energies for $R \in \{2, 3, 4, 6\}$. Keep the conductor mesh size fixed.
**c)** For $R = 4$, halve both the global and conductor mesh sizes and compare the energy changes with $E_D(4) - E_N(4)$. Why does the continuous lower bound $E_N(R) \le E_\infty$ not automatically make a computed finite element energy a rigorous lower bound?